

Introduction to Urban Data Science

CRP 4680/5680 (2025 Spring)



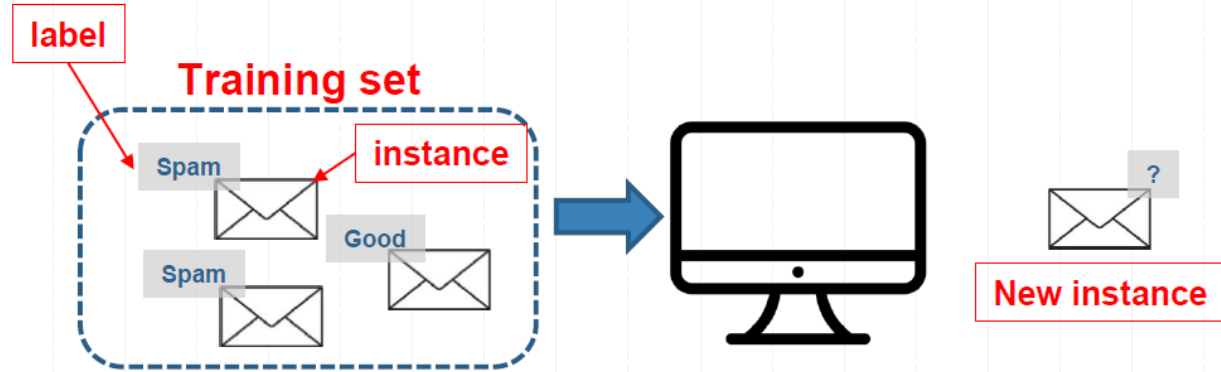
Week 14 Unsupervised Learning

Wenzheng Lee

Yujin Li

Supervised Learning

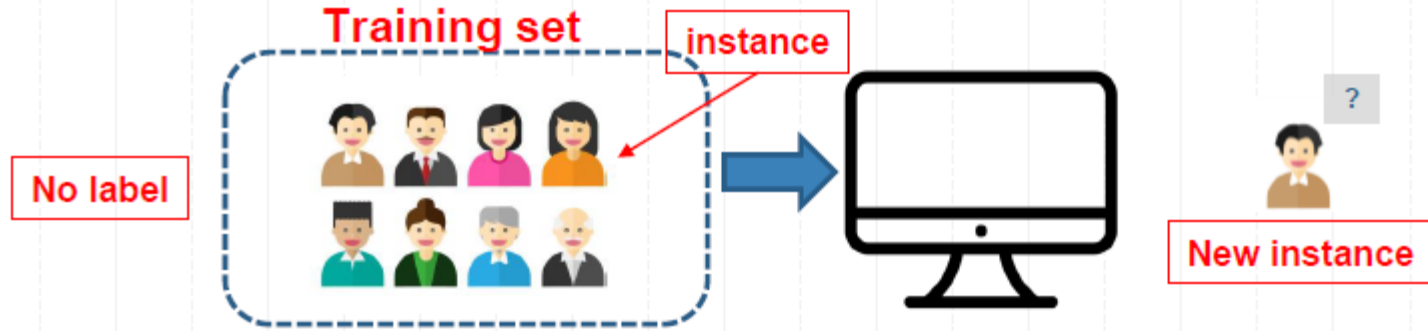
- Training



1. Supervised Learning: machine is trained with human supervision with a “teacher”, (the training set is labeled)

Unsupervised Learning

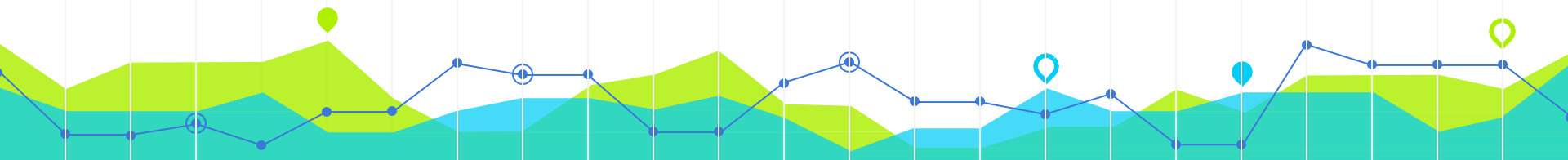
- Training



2. Unsupervised Learning: machine is trained without human supervision without a “teacher”, (the training set is not labeled)

Unsupervised Learning

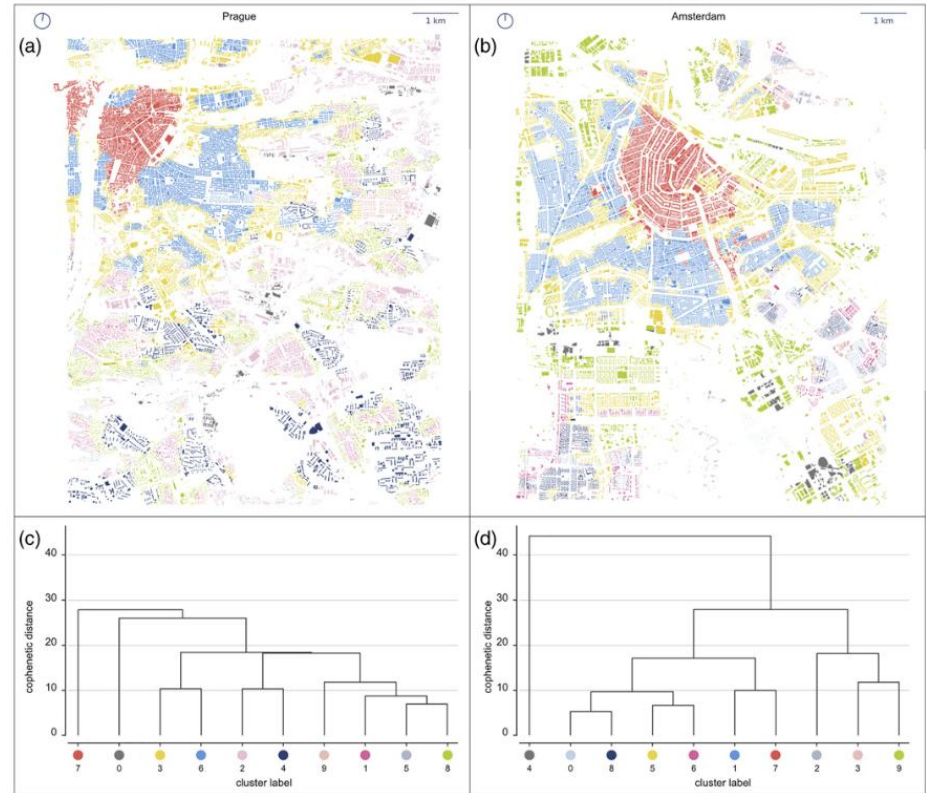
- **Clustering:** partition data into distinct groups
 - Label photos by people on iPhone
 - Recognize preference based on view/consumption history
- **Dimensionality Reduction:** Summarize a high-dimensional (many features) dataset with fewer features
 - Visualization in 2 dimension
 - Compress data, save storage space and processing time



Clustering: partition data into groups

- Spatial distribution of detected clusters in central Prague (a) and central Amsterdam (b) accompanied by dendrograms representing the results of hierarchical clustering of urban form types.
- inputs are numeric features calculated for small spatial units (e.g., urban blocks, plots, or grid cells), such as: Block size, Building density, Street network density, Building height, Compactness of parcels, Land-use mix, Street orientation entropy, Building footprint coverage

Fleischmann, M., Feliciotti, A., Romice, O., & Porta, S. (2022). Methodological foundation of a numerical taxonomy of urban form. *Environment and Planning B: Urban Analytics and City Science*, 49(4), 1283-1299.



Clustering: partition data into groups

–imagery classification



a) Sample Google Street View image and b) its corresponding semantic segmentation. Each color is mapped to a segmentation class (e.g. brown represents the "road" class). Plantation is the dominating object. Tree and vegetation have high presence for a large proportion of the images.

Clustering: partition data into groups

-imagery classification

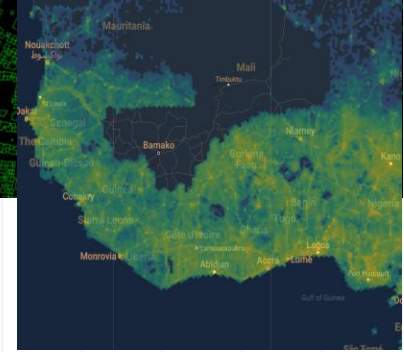


The dataset contains 1.8 billion building detections globally, covering the entire Africa continent.

Open Buildings

A dataset of building footprints to support social good applications

[Download here](#)



<https://sites.research.google/gr/open-buildings/>

Dimensionality Reduction: Summarize a high-dimensional (many features) dataset with fewer features

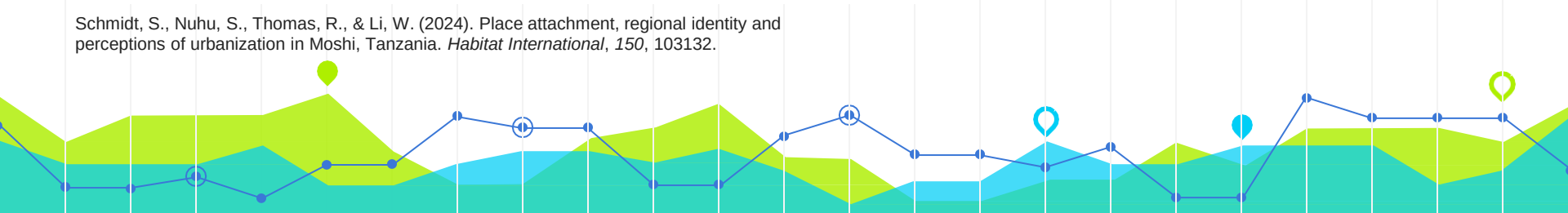
Table 4: Summary of rotated factor loadings based on PCA for 8 items on Perception of cities

	Rotated factors loadings	
	Factor 1	Factor 2
1. The city is a place with a high level of services and infrastructure.	-0.01	0.59
2. The city is a place with jobs and economic opportunity.	0.08	0.67
3. Urban way of life (<i>mazoea</i>) is a good way of life.	0.13	0.51
4. The city is a place with secure land tenure.	-0.28	0.62
5. The city is a place of chaos.	0.71	0.08
6. The city is a place of poverty.	0.65	-0.24
7. The city is a place with people who are not like me	0.68	0.07
8. Urban ways of life (<i>mazoea</i>) are not compatible with my tribal culture	0.37	0.32
Eigenvalue	1.63	1.60
% of total variation	20.23%	20.14%

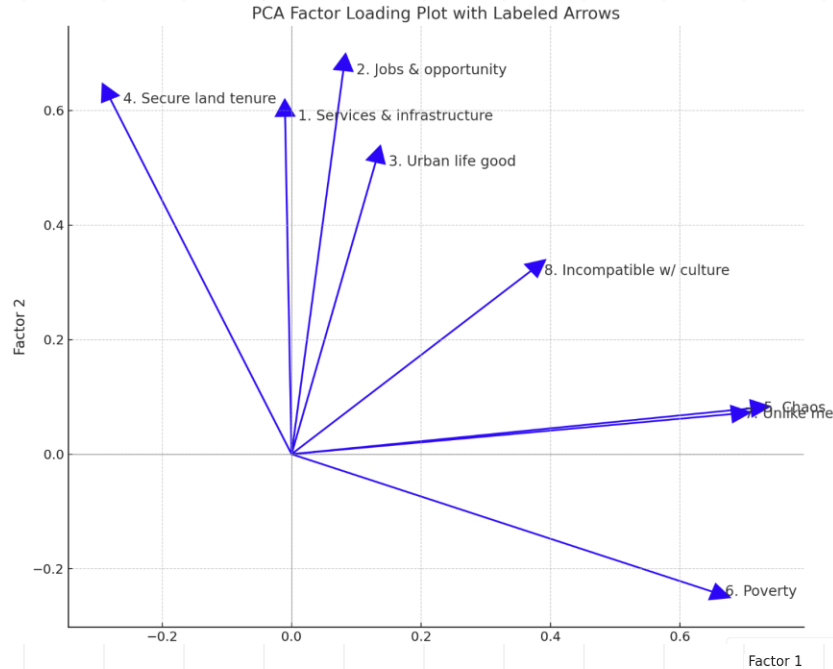
Note: Factor loadings over 0.50 appear in bold; PCA using Varimax rotation with Kaiser's criterion.

Schmidt, S., Nuhu, S., Thomas, R., & Li, W. (2024). Place attachment, regional identity and perceptions of urbanization in Moshi, Tanzania. *Habitat International*, 150, 103132.

- **Principal Component Analysis (PCA)** was used to reduce eight perception items about cities into two interpretable dimensions.
- Dimensionality reduction helps uncover underlying themes, e.g., urban opportunity and social alienation, by summarizing complex survey data into simpler factor structures.



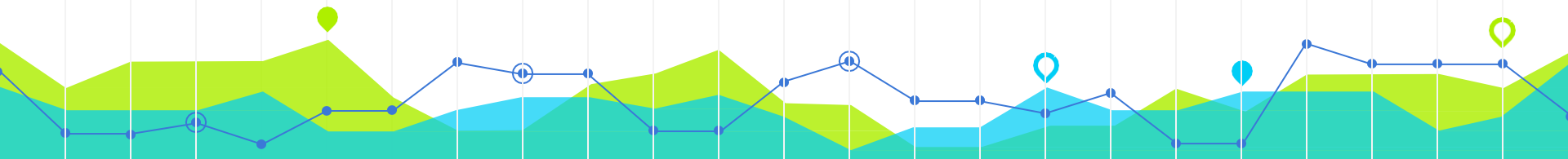
Dimensionality Reduction: Summarize a high-dimensional (many features) dataset with fewer features



- **PCA/factor analysis** helps cluster these perceptions into two interpretable themes, reducing 8 original questions into 2 factors.
- Factor 1 may reflect negative or alienating urban perceptions (e.g., chaos, poverty, exclusion). (factors 5,6, and 7)
- Factor 2 may reflect positive developmental aspects of cities (e.g., infrastructure, job opportunities, land security). (factors 1,2,3, and 4).

Unsupervised Learning

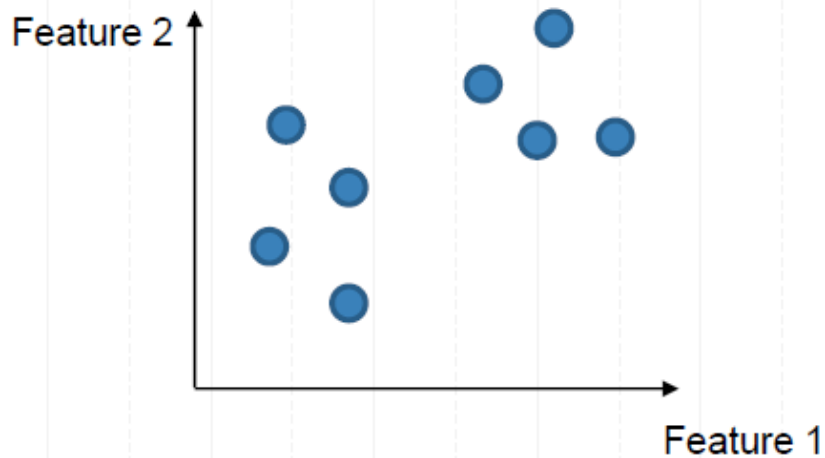
- **Clustering:** partition data into distinct groups
 - No Labels
 - How to find clusters?
 - How many clusters (k) to make?



K-Means Clustering

- **Step 1:** Choose k

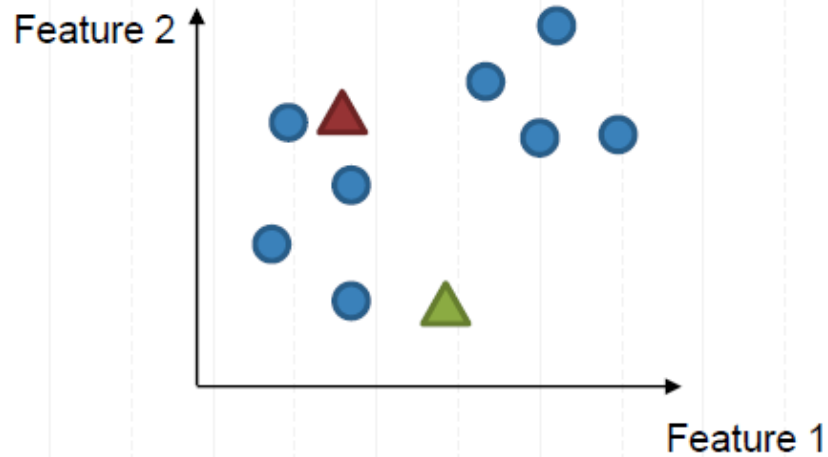
$k = 2$



K-Means Clustering

- **Step 2:** Randomly create k centroid points

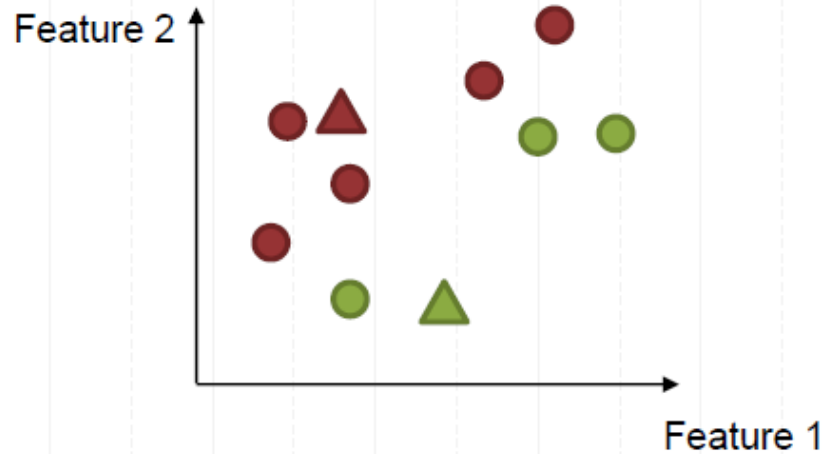
$k = 2$



K-Means Clustering

- **Step 3:** Cluster samples into closest centroid points

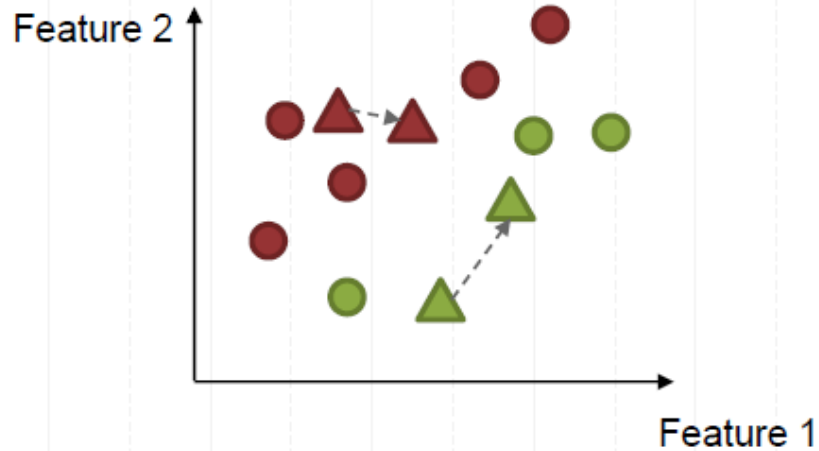
$k = 2$



K-Means Clustering

$k = 2$

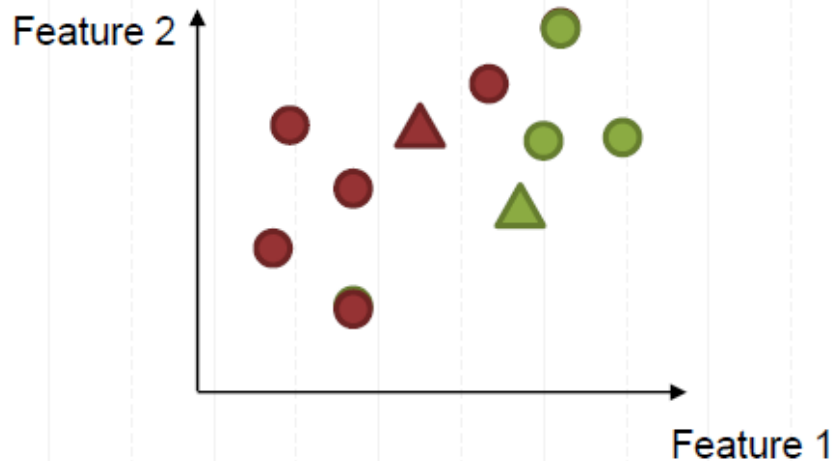
- **Step 4:** Calculate for each cluster a new centroid point based on sample features.



K-Means Clustering

$k = 2$

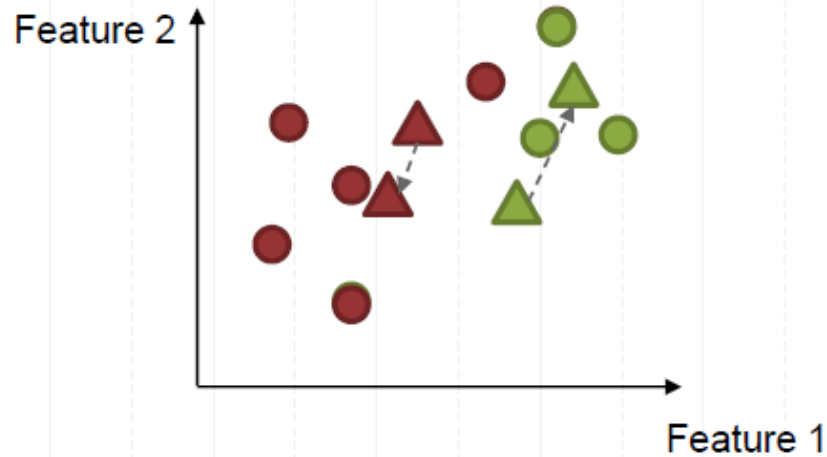
- **Step 5:** Reclassify the samples based on new centroid points



K-Means Clustering

$k = 2$

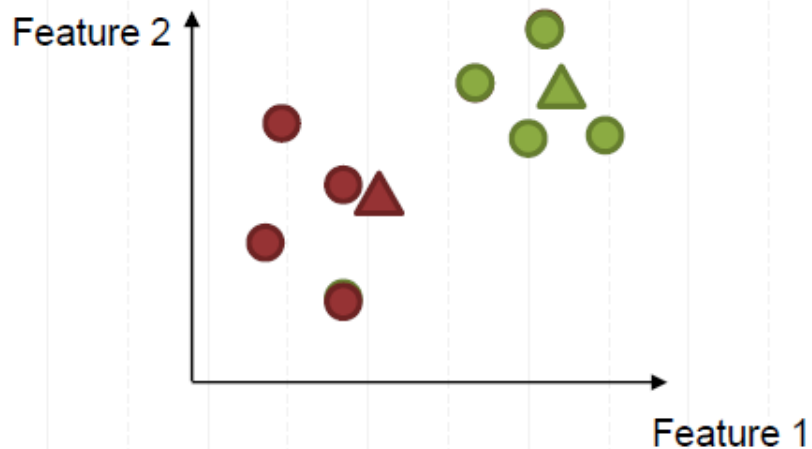
- Repeat the process until no changes in clustering



K-Means Clustering

$k = 2$

- Repeat the process until no changes in clustering



K-Means Clustering

$k = 2$

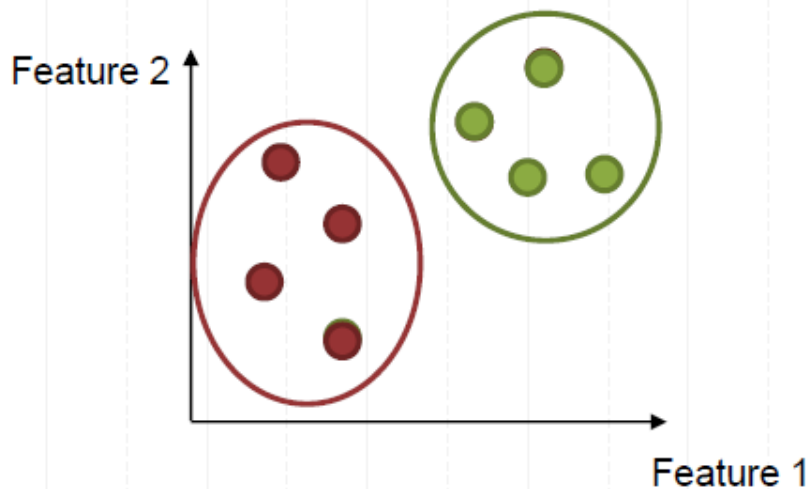
- Repeat the process until no changes in clustering



K-Means Clustering

$k = 2$

- Repeat the process until no changes in clustering



Q&A

Any questions?

You can find me at
wl563@cornell.edu

